Hilbert Space Comparison:

FFGFT $L^2(T^4)$ and UMF $\ell^2(P_N)$

An Internal Analysis of Structural Overlap and Complementarity

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Abstract

Marco Gericke (Universal Model Framework, UMF) has proposed a formal architecture in which quantum mechanics, special relativity, and fermion mass emerge from a prime-indexed Hilbert space $\ell^2(P_N)$ and its reciprocal interior $\ell^2(P_N^{-1})$, coupled by a bivector B_{OI}. This document analyses the structural relationship between UMF and FFGFT. The central result is: FFGFT's $L^2(T^4)$ is the more complete Hilbert space — it contains UMF's $\ell^2(P_N)$ as a proper subspace. However, UMF's prime-indexed structure offers genuine computational advantages in specific domains, particularly prime factorisation and number-theoretic problems. The two frameworks are therefore not in competition but stand in a relation of **mathematical containment (subset relation) and physical complementarity**: UMF's Hilbert space is a proper subspace of FFGFT's, but UMF's prime-indexed structure, bivector B_{OI}, and activation operator K_{Kriger} provide formal tools that FFGFT currently lacks.

Notation

- $P_N = \{p_1, p_2, \dots, p_N\}$ with $p_1 = 2, p_2 = 3, p_3 = 5, \dots$ (first N primes)
- $T^4 = (\mathbb{R}/2\pi\mathbb{Z})^4$ (4-torus, FFGFT state space)
- $B_{OI} = \Pi_{\text{out}} \wedge \Pi_{\text{in}} \in \Lambda^2(H_{\text{out}} \oplus H_{\text{in}})$ (UMF bivector, see §2.3)
- $\xi = 4/30000$ (FFGFT dimensionless winding parameter)
- K_{Kriger}: UMF activation operator (see §3.4)
- ξ_{UIFT}: dimensionless parameter of the UIFT framework (Doc. 190 R13), related to ξ_{FFGFT} via the bridge factor $414{,}684$

List of Symbols

Symbol	Meaning
$\xi = 4/30000$	FFGFT dimensionless winding parameter (geometrically fixed, $\kappa = 7$)
ξ_{FFGFT}	Same as ξ; used when distinguishing from ξ_{UIFT}
ξ_{UIFT}	Dimensionless parameter of UIFT: $k_B T_{\text{CMB}} \ln 2/(m_e c^2)$ (Doc. 190 R13)
$L^2(T^4)$	Hilbert space of square-integrable functions on the 4-torus T^4
T^4	4-dimensional torus $(\mathbb{R}/2\pi\mathbb{Z})^4$; FFGFT state space
$\ell^2(P_N)$	UMF Hilbert space: square-summable sequences over first N primes
P_N	Set of first N prime numbers $\{2, 3, 5, \ldots, p_N\}$
$w_p = \ln p/\sqrt{p}$	Prime weight in UMF inner product
B_{OI}	UMF bivector: $\Pi_{\text{out}} \wedge \Pi_{\text{in}}$; couples the two spaces
Π_{out}	UMF Prime Boundary $= \ell^2(P_N)$ (exterior, metric domain)
Π_{in}	UMF Reciprocal Interior $= \ell^2(P_N^{-1}) \oplus \ell^2(Q_{ZN})$ (mass domain)
K_{Kriger}	UMF activation operator; determines which interface information generates curvature
$\chi(x)$	Dimensionless activation parameter: $k_B T(x) \ln 2 \cdot \tau_t(x)/(\hbar/4)$
$\chi = 1$	Landauer threshold: boundary between inactive and active structure
D_P	UMF chiral Dirac operator on $\ell^2(P_N) \otimes \mathbb{C}^2$
$\tau \cdot \mu$	Coherence-time – information-mass product; $= \hbar/c^2$ (both frameworks)
$\varepsilon \approx 0.001$	FFGFT non-closure of T^4; generative structural residual
θ_4	Continuous fourth-torus phase coordinate; absent from $\ell^2(P_N)$
n_θ, n_ϕ	FFGFT winding numbers in two independent torus directions
ι	Embedding $\ell^2(P_N) \hookrightarrow L^2(T^4)$ (containment map)

1 The Two Hilbert Spaces

1.1 FFGFT: $L^2(T^4)$

FFGFT's state space is $L^2(T^4)$ — the space of square-integrable functions on the 4D torus $T^4 = (\mathbb{R}/2\pi\mathbb{Z})^4$. The natural orthonormal basis consists of Fourier modes:

$$e_{\mathbf{n}}(\theta) = \frac{1}{(2\pi)^2} e^{i(n_1\theta_1+n_2\theta_2+n_3\theta_3+n_4\theta_4)}, \qquad \mathbf{n} \in \mathbb{Z}^4 \tag{244.1}$$

This space is:

- **Complete**: every Cauchy sequence converges in $L^2(T^4)$
- **Separable**: the Fourier basis $\{e_{\mathbf{n}}\}_{\mathbf{n}\in\mathbb{Z}^4}$ is countable and dense
- **Continuous**: admits continuous winding numbers and phase structure
- **Indexed by** $\mathbb{Z}^4$: all integer winding number combinations, including prime-indexed ones as a special case

The single parameter $\xi = 4/30000$ determines which winding modes are dynamically preferred — but all modes are structurally present in $L^2(T^4)$.

1.2 UMF: $\ell^2(P_N)$

UMF's primary space is $\ell^2(P_N)$ — the space of square-summable sequences indexed by the first N prime numbers, with inner product:

$$\langle f|g\rangle_{P_N} = \sum_{p\le p_N} w_p \, \overline{f(p)}\, g(p), \qquad w_p = \frac{\ln p}{\sqrt{p}} \tag{244.2}$$

This space is:

- **Complete** for fixed finite N: $\ell^2(P_N) \cong \mathbb{C}^N$
- **Discrete**: no continuous phase structure
- **Indexed by primes only**: $\{2, 3, 5, 7, 11, \dots, p_N\}$
- **Weight-divergent** as $N \to \infty$: $\sum_p w_p = \sum_p \ln p/\sqrt{p}$ diverges (by comparison with $\sum 1/\sqrt{p}$)

2 Containment Relation

2.1 $\ell^2(P_N)$ is a Subspace of $L^2(T^4)$

The prime-indexed basis vectors of UMF correspond to specific Fourier modes of $L^2(T^4)$. Consider the embedding:

$$\iota : \ell^2(P_N) \hookrightarrow L^2(T^4), \qquad f(p) \mapsto f(p) \cdot e_{(p,0,0,0)}(\theta) \tag{244.3}$$

This maps each prime-indexed coefficient to a Fourier mode with winding number $(p, 0, 0, 0)$ in the first torus direction. The embedding is isometric up to normalisation and injective. Therefore:

$$\boxed{\ell^2(P_N) \hookrightarrow L^2(T^4), \quad \text{proper subspace}} \tag{244.4}$$

$L^2(T^4)$ contains far more: all integer winding numbers in all four directions, all non-prime modes, all mixed-direction modes, and the continuous phase structure θ_4 which is absent from $\ell^2(P_N)$ entirely.

2.2 What UMF's Space Cannot Represent

The following FFGFT structures have no counterpart in $\ell^2(P_N)$:

FFGFT Structure	In $L^2(T^4)$	In $\ell^2(P_N)$
Continuous phase θ_4	✓ (full range)	$\times$ (absent)
Non-prime winding modes	✓	$\times$
Mixed-direction winding (n_1, n_2, n_3, n_4)	✓	$\times$
Non-closure ε (winding residual)	✓	$\times$
Z3^3 topological constraint	✓	partially
4π spinor periodicity	✓	$\times$
Lorentz structure (from T^4 geometry)	✓	derived via RG

2.3 The Missing Object: UMF Bivector B_{OI}

Marco Gericke's central formal contribution is not $\ell^2(P_N)$ alone but the wedge product of the two spaces:

$$B_{OI} = \Pi_{\text{out}} \wedge \Pi_{\text{in}} \;\in\; \Lambda^2(H_{\text{out}} \oplus H_{\text{in}}) \tag{244.4b}$$

where $\Pi_{\text{out}} = \ell^2(P_N)$ and $\Pi_{\text{in}} = \ell^2(P_N^{-1}) \oplus \ell^2(Q_{ZN})$.

B_{OI} has three crucial properties:

1. **Projection-stable**: neither π_1 (metric projection) nor π_2 (spinor/mass projection) alone contains it, but both projections preserve its existence as a structural invariant.
2. **Generates observable geometry**: $g_{\mu\nu}^{\text{phys}} = g_{\mu\nu}^{(0)} + \lambda_B \, B_{\mu\alpha}^{\text{act}} B_{\nu}^{\text{act}\,\alpha}$
3. **Not derivable from either space alone**: it is the coupling between Π_{out} and Π_{in}, not a state within either.

This is the formal realisation of what Stefaan Vossen called the GIA projection-stable invariant, and what FFGFT described verbally as "mutual constitution". B_{OI} is *not* an element of $L^2(T^4)$ — it is a relation *between* spaces. The containment $\ell^2(P_N) \hookrightarrow L^2(T^4)$ holds for the spaces themselves; B_{OI} is a genuinely new object that FFGFT currently has no algebraic counterpart for.

3 Where UMF Offers Genuine Added Value

The containment relation does not make UMF redundant. Three domains where the prime-indexed structure offers advantages:

3.1 Prime Factorisation and Number Theory (Doc 075, 076)

FFGFT Doc 075 treats RSA factorisation via T0-field dynamics — period-finding as wave propagation rather than quantum superposition. The natural computational basis for this problem is prime-indexed. UMF's $\ell^2(P_N)$ with the prime-weighted inner product $w_p = \ln p/\sqrt{p}$ is the *natural* Hilbert space for number-theoretic computations: the weights encode the prime number theorem density $\pi(x) \sim x/\ln x$ directly.

In $L^2(T^4)$, prime modes are present but not distinguished. In $\ell^2(P_N)$, they are the entire basis. For factorisation problems, UMF's representation is computationally more efficient.

3.2 The Dirac Operator D_P as Computable Spectrum

UMF's chiral Dirac operator on $\ell^2(P_N) \otimes \mathbb{C}^2$:

$$D_P = \begin{pmatrix} 0 & A \\ A^\dagger & 0 \end{pmatrix}, \quad A_{pq} = w_p \, \delta_{pq} \tag{244.5}$$

has a discrete, explicitly computable spectrum. The spectrum of the corresponding operator on $L^2(T^4)$ requires solving the full torus eigenvalue problem — substantially harder. For mass-spectrum calculations in the lepton sector, both approaches give $\sim 1\%$ agreement with PDG, but UMF's computation is more direct.

3.3 Independent Confirmation of Lorentz Invariance

UMF derives Lorentz invariance as the IR fixed point of the anisotropy RG flow $u(\mu) \to u^* = 1$. FFGFT derives it from the T^4 winding geometry and the constraint $\tilde{T} \cdot m = 1$. These are structurally independent derivations arriving at the same result. The independence strengthens both: if two different mathematical starting points require Lorentz invariance, the result is more robust.

3.4 The Activation Operator $K_{\textsf{Kriger}}$

UMF defines the dimensionless activation parameter:

$$\chi(x) = \frac{k_B T(x) \ln 2 \cdot \tau_t(x)}{\hbar/4} \tag{244.6a}$$

and the soft activation gate:

$$A(x) = \frac{1}{1 + e^{-s(\chi(x)-1)}} \tag{244.6b}$$

Physical geometry receives contributions only from the activated fraction:

$$\Phi_{\text{grav}} = A(x) \cdot \Phi_{\text{interface}} \tag{244.6c}$$

Key insight: The threshold $\chi = 1$ is exactly the Landauer bound. For $\chi < 1$, the bivector B_{OI} exists mathematically but generates no curvature — it is structurally present but physically inactive. This formalises what FFGFT has described qualitatively as the boundary between geometrically present and observationally active structure. FFGFT has no algebraic analogue of K_{Kriger}. It is UMF's genuine formal contribution, independent of the containment relation.

4 The $\tau \cdot \mu$ Identity: Independent Derivation

Both frameworks arrive at the same identity independently:

$$\tau \cdot \mu = \frac{\hbar}{c^2} \tag{244.7}$$

where $\tau = \hbar/(k_B T_{\text{CMB}} \ln 2)$ and $\mu = k_B T_{\text{CMB}} \ln 2/c^2$.

In FFGFT this is the coherence time – information-mass product at the CMB scale. In UMF this is the statement that the Kriger activation threshold, evaluated at T_{CMB}, equals the Compton-scale information mass. Two different derivational paths, identical result. The agreement is not coincidental: it reflects the bridge formula $\xi_{\text{FFGFT}}/\xi_{\text{UIFT}} \approx 414{,}684$ (Doc 190 R13), which connects the same CMB temperature to both frameworks through different normalisations.

5 The Non-Redundancy Principle

The most useful framework is not the most complete one but the one that derives the most from the fewest assumptions. By this criterion:

Table 1 — Technical comparison:

Criterion	FFGFT	UMF
Free parameters	1 (ξ)	5 (Distinction Logic axioms + N)
Hilbert space	$L^2(T^4)$ (complete)	$\ell^2(P_N) \subset L^2(T^4)$
Lorentz derivation	from T^4 geometry	from RG flow (independent)
Mass spectrum	from winding recursion	from D_P spectrum (compact)
Activation boundary	qualitative	K_{Kriger} (explicit)
Factorisation	wave propagation	natural prime basis

Table 2 — Ontological / structural comparison:

Criterion	FFGFT	UMF
Primitive	Geometry (T^4)	Distinction Logic
Bivector B_{OI}	not present	$\Pi_{\text{out}} \wedge \Pi_{\text{in}}$
"Mutual constitution"	described verbally	algebraically realised
Ontological commitment	geometric primacy	distinction-logic primacy

The two frameworks occupy different positions in the projection space $\mathcal{M}$ (Stefaan Vossen's comparative architecture): they share projection-stable invariants (Lorentz structure, $\tau \cdot \mu$, mass spectrum) while differing in generative ontology and computational representation.

6 Conclusion: Complementarity without Redundancy

1. $\ell^2(P_N) \hookrightarrow L^2(T^4)$: UMF's Hilbert space is a proper subspace of FFGFT's. $L^2(T^4)$ is the more complete structure.
2. UMF is not redundant: it offers computational advantages in prime-indexed domains, an explicit activation operator, an independent derivation of Lorentz invariance, and a computable Dirac spectrum.
3. The $\tau \cdot \mu$ identity confirms structural alignment at the CMB scale, consistent with Doc 190 R13.
4. The appropriate response to Marco Gericke's mail:
   (a) Acknowledge that UMF's $\ell^2(P_N)$ is a proper subspace of FFGFT's $L^2(T^4)$ — this is mathematics, not a value judgement.
   (b) Accept that UMF's B_{OI} and K_{Kriger} are formal objects FFGFT lacks; they are relations *between* spaces, not states within one space, and are therefore not contained in $L^2(T^4)$.
   (c) Propose a joint short note: mapping ξ_{FFGFT} to the UMF compactification radius R_χ via the bridge constant $\xi_{\text{FFGFT}}/\xi_{\text{UIFT}} \approx 414{,}684$ (Doc 190 R13, where ξ_{UIFT} is the dimensionless parameter of the UIFT framework), and invite Marco to complete the UMF entry in Stefaan's comparative schema (§5.x).
5. No ontological priority claim is needed: the containment relation is mathematical, not ontological. UMF may generate $\ell^2(P_N)$ from distinction logic;

FFGFT generates $L^2(T^4)$ from geometry. The subspace relation holds regardless of which generative story one prefers.